FGCU QUANTUM HACKATHON · IBM QISKIT FALL FEST 2026

Rewind & Amplify

Grover’s algorithm inside a quantum recurrent state machine



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One click record: 64 steps, 5 clicks

Rare events hold the answers, and cost the most to see

Why it matters

Tiny hidden defects limit today’s qubits, and nobody can put a probe on one. The evidence is in rare patterns of clicks from one qubit’s readout.

Who it hits

The engineers who build and calibrate quantum chips, and anyone paying for quantum time by the second.

The catch

To study what happens around a rare event, you pay for every run in which it didn’t happen.

27.5 million

runs of our model to match one real 64-step click record, once

Median over 4,096 held-out records, from the exact filter

The real-world problems this work is for

Hidden defects make chips unreliable

Microscopic two-level defects drift in and out of resonance with qubits. They are the leading explanation for the largest swings in qubit lifetime.

This work: learns a defect from one qubit’s clicks, with no probe. Shown in simulation.

Calibration data can miss them

Compilers choose qubits from reported error rates. On ibm_marrakesh, the qubits ours chose showed **1.5% to 3.4%** error per gate, against about 0.2% reported.

This work: a short echo circuit caught it, on a real chip.

Error correction has to find them first

Fault-tolerant codes can quarantine a bad qubit, whether it came from fabrication or appears mid-run. But something has to detect it.

This work: a candidate detector from rare click patterns. Not yet tested inside a code.

Studying rare events is expensive

IBM charges **$96 a minute** on pay-as-you-go. Matching one 64-step record by brute force takes at least 2 hours: about **$12,000**.

This work: 125 machine calls with steering and Grover, on future hardware.

Beyond computers

Single-photon sources for quantum networks blink as charge traps switch. Labs study this through click correlations, the same kind of signal we read.

Where we stand

Defects found in simulation, and circuits tested on two real chips. We have not yet read a real defect from a real chip’s clicks.

Klimov 2018; Carroll 2021; Siegel 2023; Berhane 2017; IBM pricing, Nov 2026. Cost: 27.5 million shots at our measured 283 µs a shot, a lower bound.

Our model: a quantum recurrent state machine

Drive
The emitter is pushed a little
$RY(\theta)$

Detune
The hidden defect turns
$RZ(\zeta)$

Exchange
They swap energy
ψ

Emit
The emitter may give light
γ

Read, then reset
Click or no click: the only thing we see
$|0\rangle$

Repeat every step. Two hidden qubits hold the memory, one bit comes out, and a step costs 4 two-qubit gates on IBM Heron.

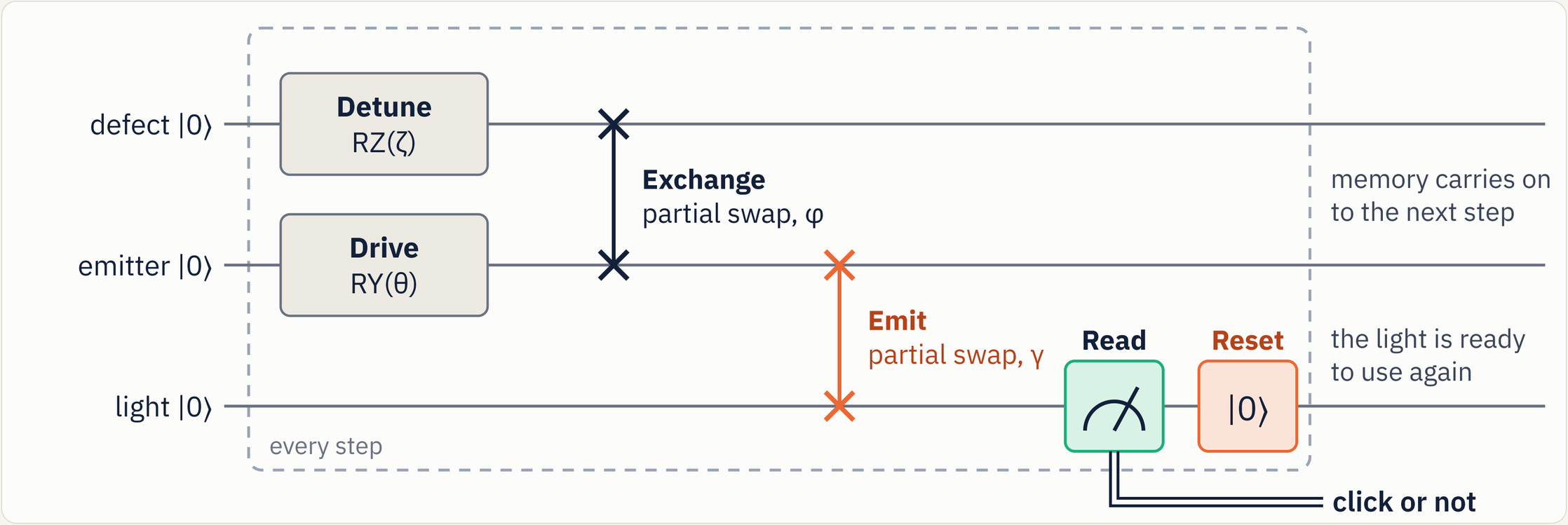
Our work earlier this week

≤ 0.1%
prediction error left when it learns a defect from clicks alone

17%
left by a tuned LSTM on the same records

But
it only generates. You can’t ask it a question, only run it and discard.

One step of the hybrid QRSM, as a circuit



The memory

Emitter and defect stay coherent and are never measured. They carry the state from step to step.

The partial swaps

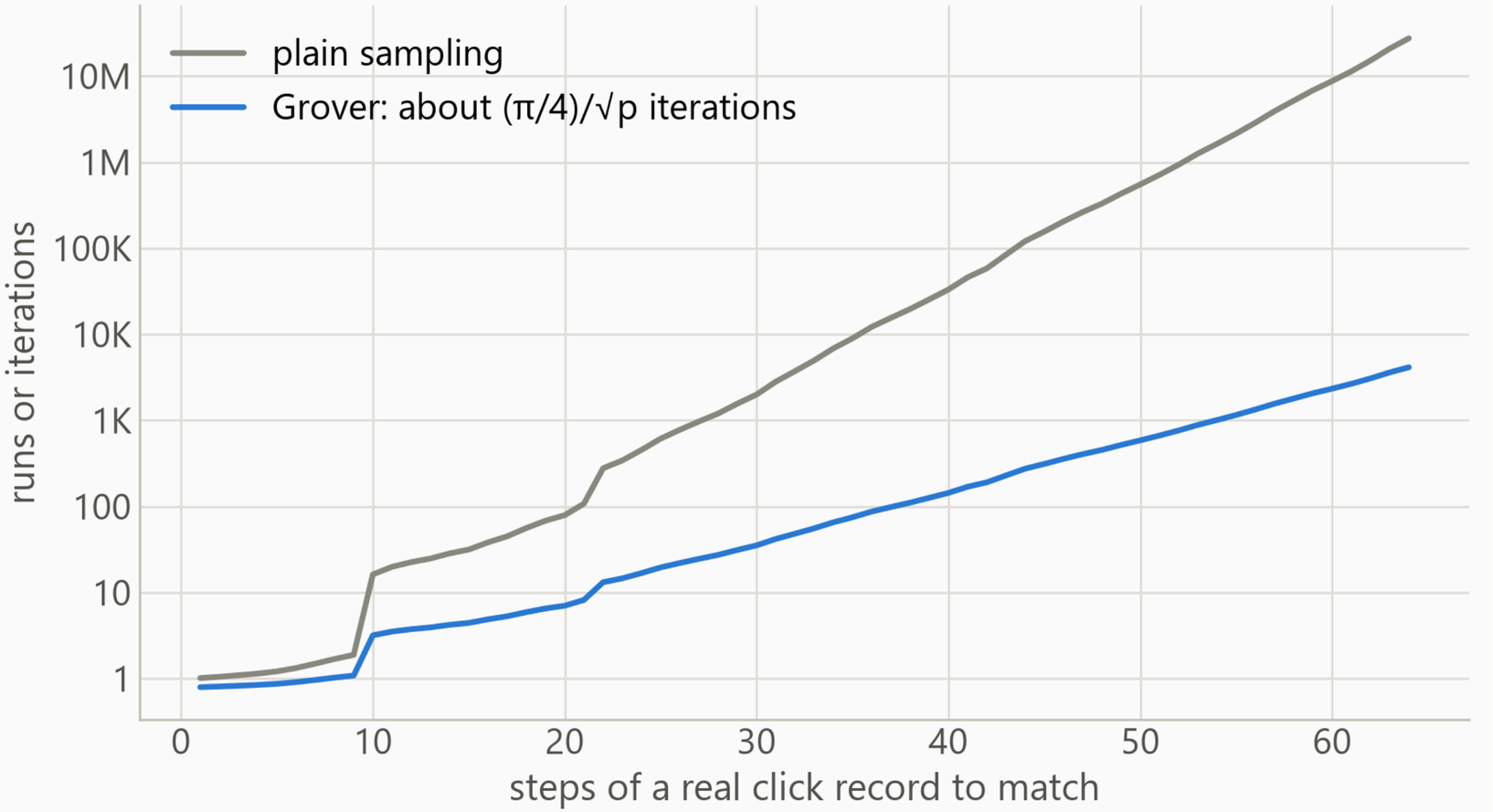
Exchange and Emit each move part of an excitation. A step costs 4 two-qubit gates on IBM Heron.

What makes it hybrid

The light is read and reset, so information leaves and the memory can forget. Keep an eye on that reset.

Redrawn from the team's QRSM circuit diagrams (v5 4), with one defect. Defaults: θ = 0.6, γ = 0.2, φ = 0.45, ζ = 0.3.

Asking it a question means throwing runs away



To condition on a record, you keep only the runs that match it.

The cost is **1/p**, and every step of the record multiplies it.

Grover's algorithm offers **1/√p**. If it can be made to work here.

Median over 4,096 held-out records of the hidden-defect machine, scored by the exact filter

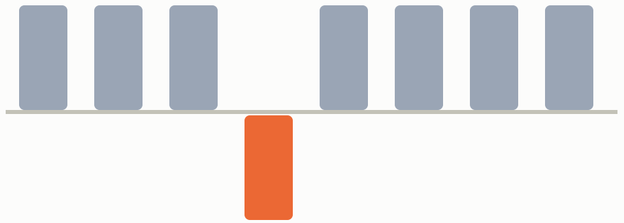
Grover’s algorithm in one minute

1. Start



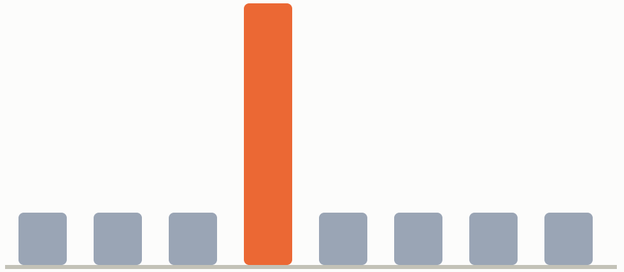
Every answer is held at once, with equal weight.

2. Mark



Flip the sign of the answers you want.

3. Reflect

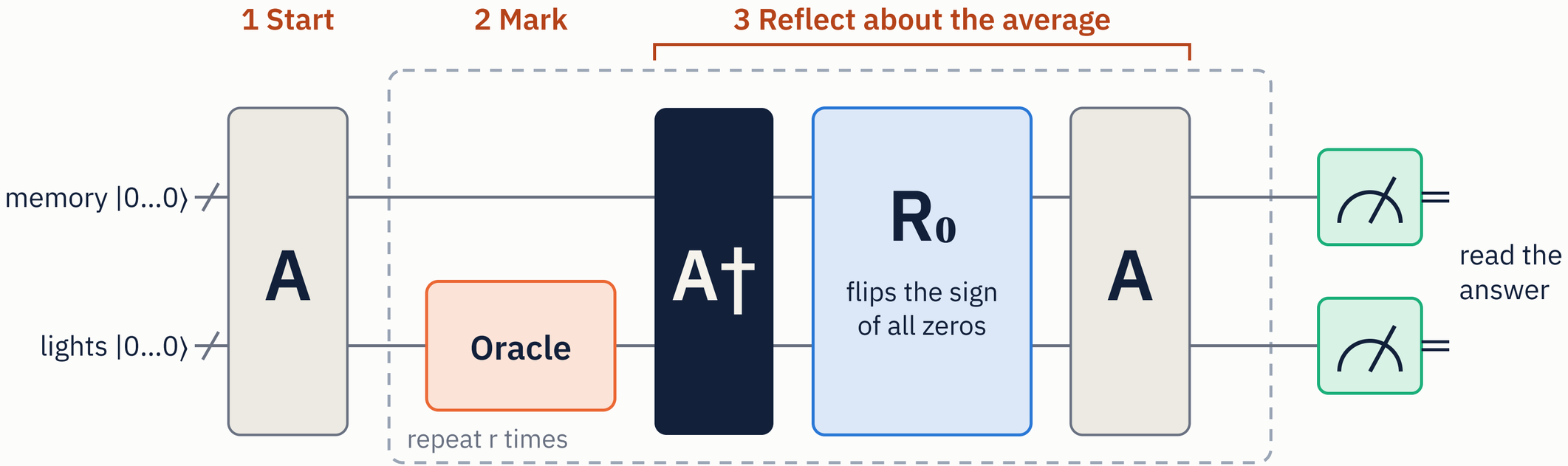


Flip everything about the average. The marked answer grows.

Repeat. A **10%** chance becomes **99.9%** in two rounds. Searching N items takes about $\sqrt{\mathbf{N}}$ rounds, where sampling takes N.

$$P = \sin^2((2r + 1)\,\theta)$$
$$p = \sin^2\theta$$

Grover’s algorithm, as a circuit



A
Prepares every answer at once. Textbook A is a row of Hadamards. Ours will be the whole machine.

Oracle
Flips the sign of the answers we want. For a click at one step, that’s a single Z gate.

A†, R₀, A
Rewind, flip the sign of all zeros, replay. R₀ is X gates, a multi-controlled Z, then X gates.

Q = A R₀ A† O
One iteration, two machine calls. Note the A†: the machine has to run backwards.

In Qiskit: grover_operator(oracle, state_preparation=A). The resilient build keeps this shape, with one angle changed and a 3-qubit R₀.

You can’t Grover what you can’t rewind.

A reset can’t be undone

Each Grover round runs the machine backwards. Ours resets a qubit every step.

A state isn’t a database

The hidden memory is compressed. It doesn’t hold records you can address.

Fewer queries isn’t less time

Rewinding and reflecting cost gates, and on real chips gates cost errors.

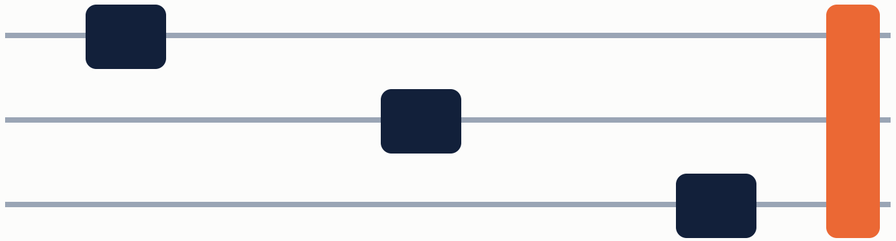
Move 1: Purify. A fresh qubit replaces every reset

Before: one light, reset each step



Measure and reset: cheap in qubits, but the past is gone.

After: a fresh light per step



Nothing measured until the end: one reversible circuit, A.

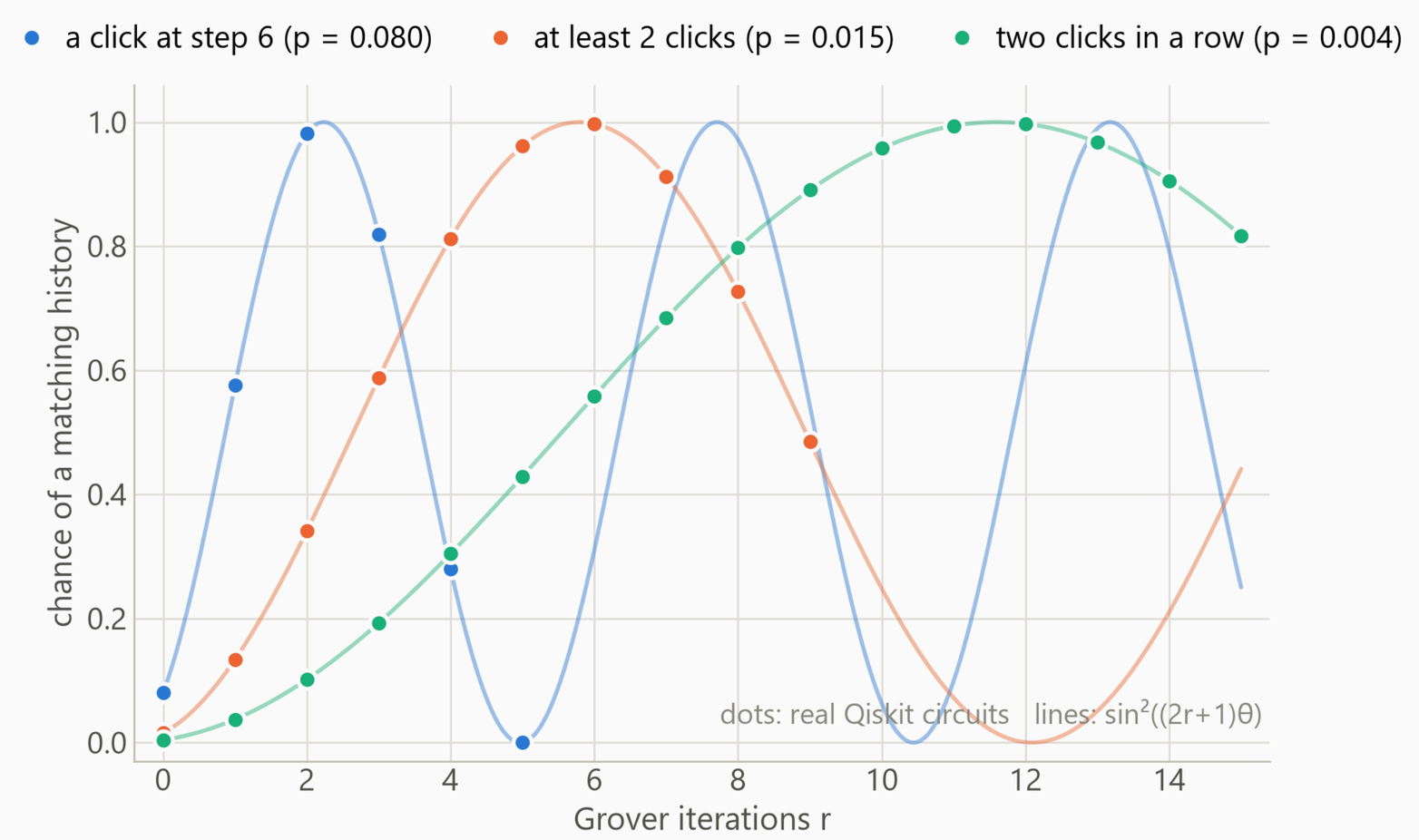
$$A|0\rangle = \Sigma \text{ over } y \text{ of } |m_y\rangle\ |y\rangle$$

The state holds **every click history at once**, each with weight $\sqrt{P(y)}$. Grover's search space is the machine's own past.

4×10^{-16} largest amplitude error against the exact model

$p = 0.61$ reset and purified circuits click alike, 1,000,000 shots each

Grover now searches the machine’s histories



8×10^{-15}

largest gap from theory, over six questions

Exact

conditioning: matches keep their true relative odds, hidden qubits included

725×

fewer machine calls at p = 10^{-6}: slope −0.50 against −1.00

Qiskit grover_operator on the purified QRSM, T = 6, noiseless. Each iteration counts as two machine calls.

Why a partly coherent, partly reversible QRSM

How coherent? The middle is what works

Fully coherent

Nothing is read or reset, so it can’t forget.

51%

on “ends with A”: chance

Hybrid: ours

Memory stays quantum, and a fresh qubit carries information away.

100%

on the same task

Fully measured

Every qubit is read each step: a classical automaton.

Classical

settles to uniform

How reversible? Only as far as memory reaches

Forward only

Read and reset: no way back, so Grover can’t run.

1 qubit

for the light, at any length

Full rewind

Purify: the whole run can be reversed.

T qubits

one fresh light per step

Bounded rewind

It forgets, so rewind only the recent past.

8 steps

held 0.80 to 0.87 out to T = 18

Where three fields meet

Probability

Every click record gets a probability: a hidden Markov model with quantum memory. No classical HMM up to 32 states matched one qubit on a driven atom.

Data science

A recurrent sequence model that can forget. Three physical numbers left 0.1% of the error, where a tuned LSTM with up to 67,713 left 17%.

Quantum mechanics

Each click updates the memory through Kraus operators: an open quantum system. Any open system is a reversible one with more qubits, which is why purifying works.

“Ends with A” and the driven atom: our earlier tests (v1 07, 08, 12). Bounded rewinding: noiseless only, and it distorts history older than its horizon.

Noise punishes depth, so we redesigned around it

456

two-qubit gates: textbook, T = 5

Grover is famously fragile. Each iteration rewinds the whole history and checks every qubit. At 0.2% error per gate, only about 40% of those runs finish clean.

The resilient architecture: four moves

1 Purify

Fresh qubits replace resets, so the machine can be rewound.

Price: one qubit per step

2 Steer

Tune one gate so the event is common. Free, and exact.

4,400× fewer runs on a real record

3 Amplify locally

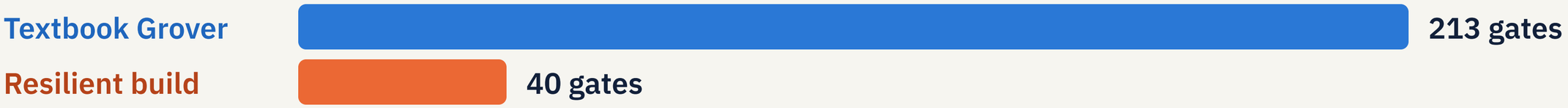
One iteration, with a reflection on 3 qubits at any length.

With steering: 3 to 5× fewer gates

4 Verify

The click is measured, so a failed shot shows itself and is dropped.

On a real chip: 87% of shots kept



Bars: two-qubit gates for one rare click at T = 3, compiled for ibm_kingston.

Move 2: Steer. Part of the rarity is free to remove

A click is rare for two reasons

Did it emit? The emission factor

Set by one gate angle. Turning it up scales every matching history by the same constant, so the answer stays **exact**.

Was it excited? The population factor

No gate setting removes this. It is Grover’s real job.

Tune p to exactly **1/4** and **one** iteration gives certainty, where textbook needs two or three.

Machine calls per conditioned sample of a real 64-step record

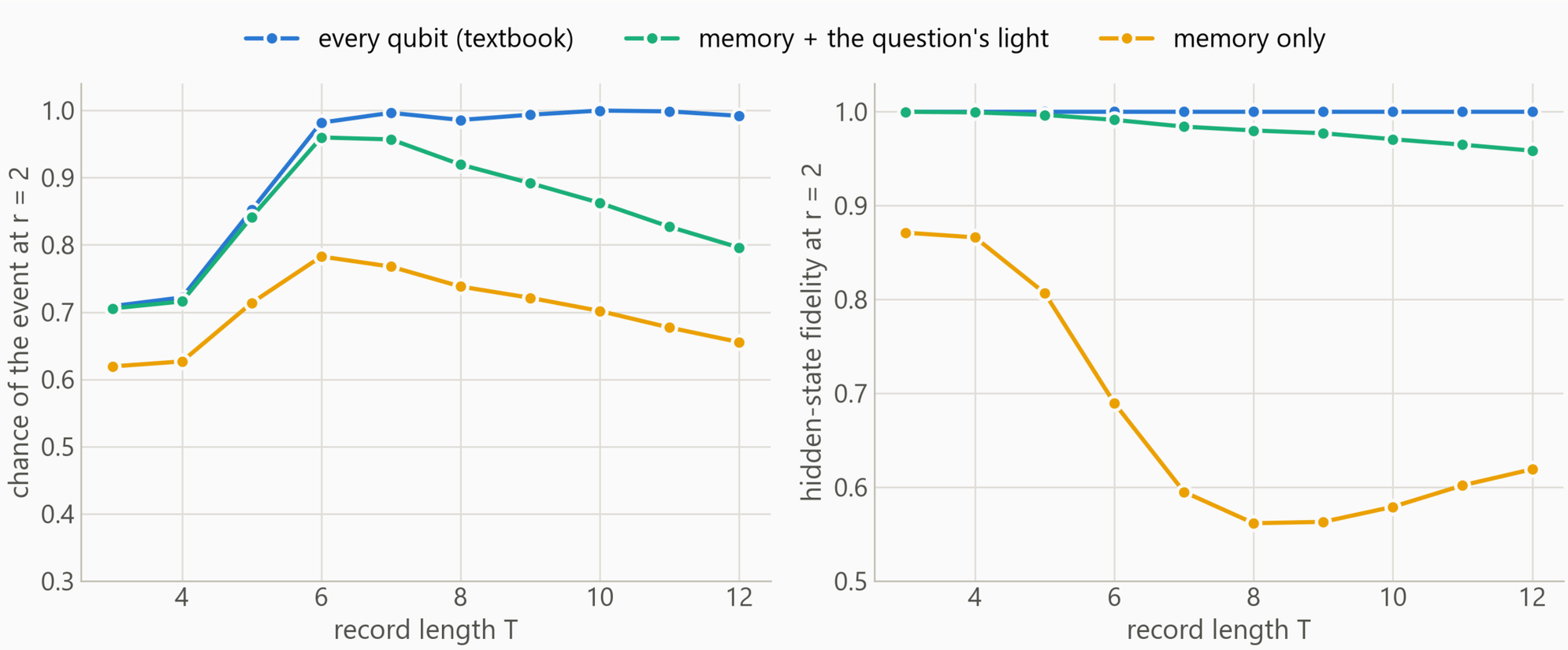
27,495,891
plain sampling

6,240
steered: 4,400× fewer, classical maths

125
steered, then Grover: 50× fewer again

Median of 4,096 held-out records, 5 clicks each. Steering sets every clicked step to emit with certainty. Exact for clicks at known steps.

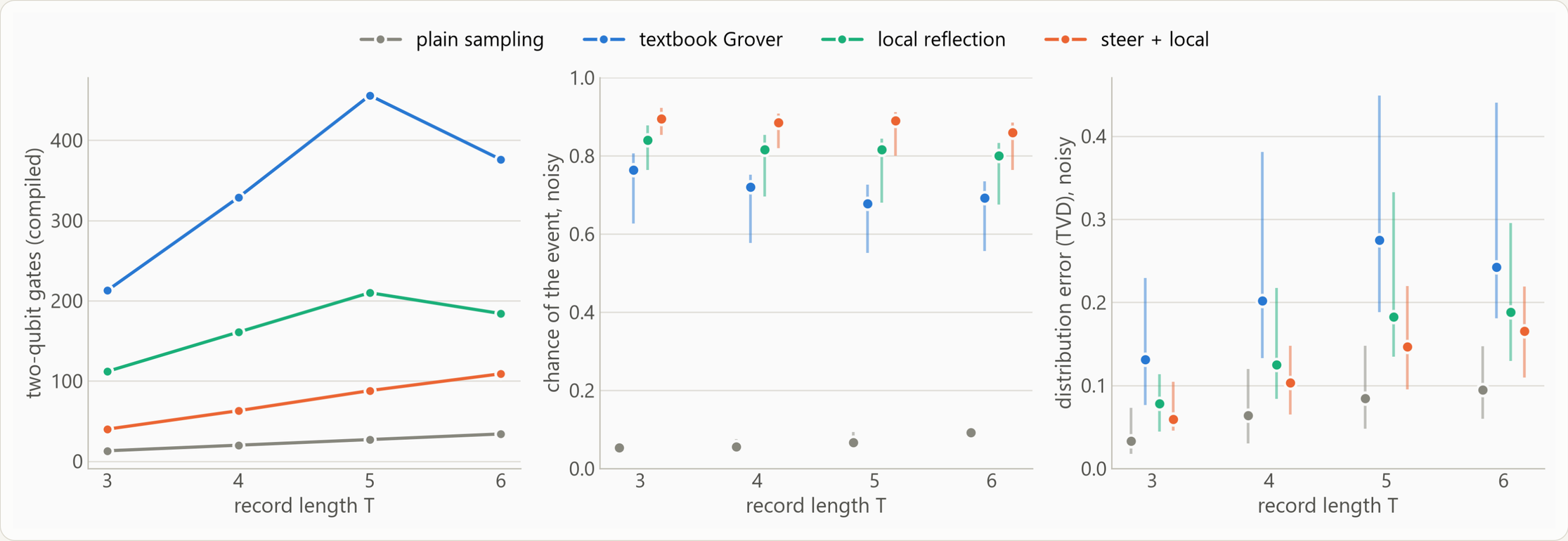
Move 3: Reflect only where the question looks



Watching only the memory looked great on hit rate, then failed: it scrambled the hidden state. Our own test caught it.

Adding the question's light fixes it: **3 qubits** at any length, in place of every qubit. It drifts on long records.

Across five IBM device models: fewer gates, better answers



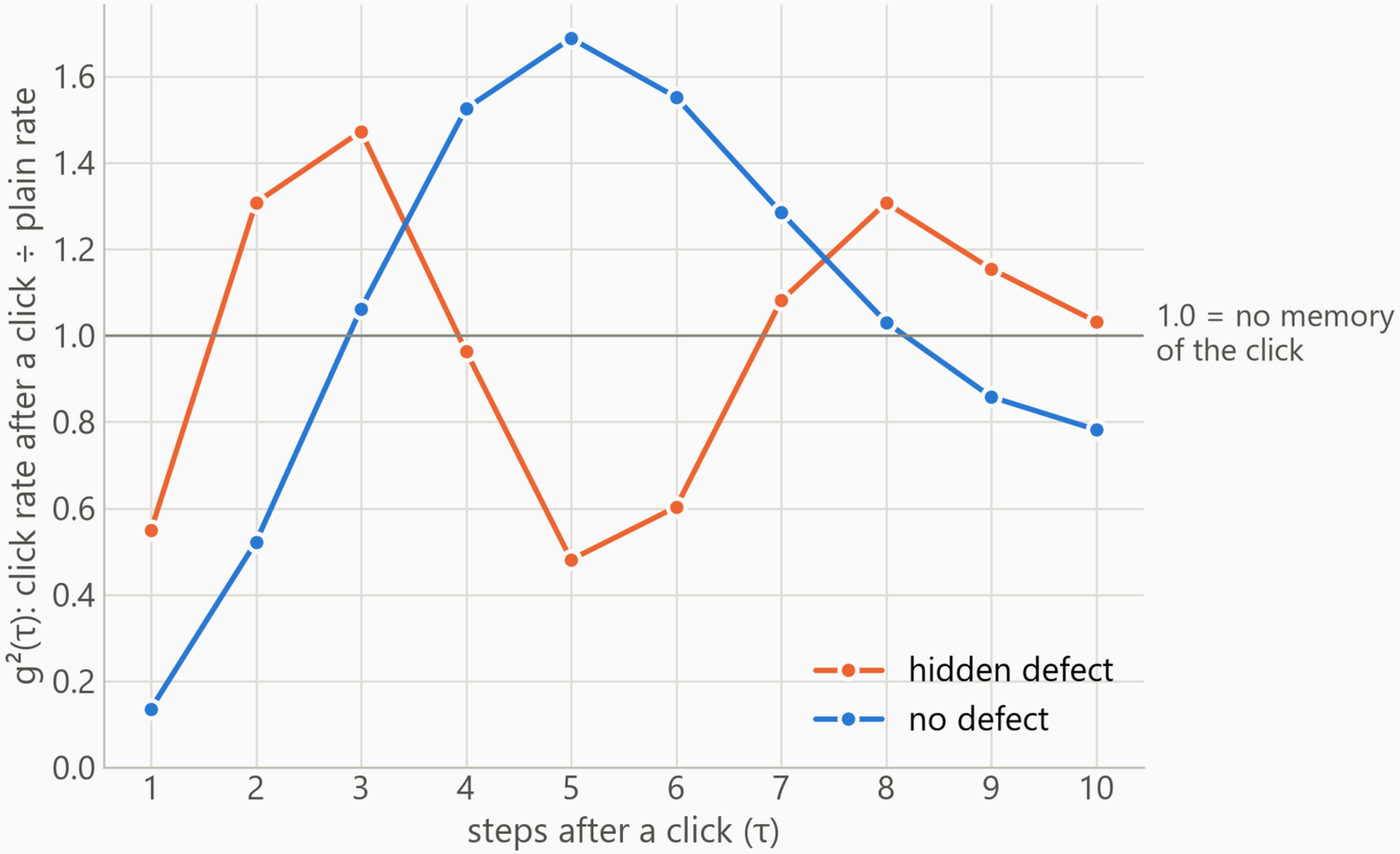
3–5× fewer two-qubit gates than textbook Grover

0.88 chance of the event, against 0.71 for textbook

0.12 distribution error, against 0.21 for textbook

Better on both in all 20 device and length cells. Dots: median; bars: range. Plain sampling keeps only 5–9% of shots.

Now it can answer a physics question



What happens right after a rare click?

With no defect, a click **wipes the machine**: what follows simply restarts the run.

A hidden defect **remembers across the click**. This curve is a fingerprint of hidden memory.

2.4σ → 7.1σ

telling defect from no defect at 4,096 shots on noisy device models, once we amplify the click and keep running

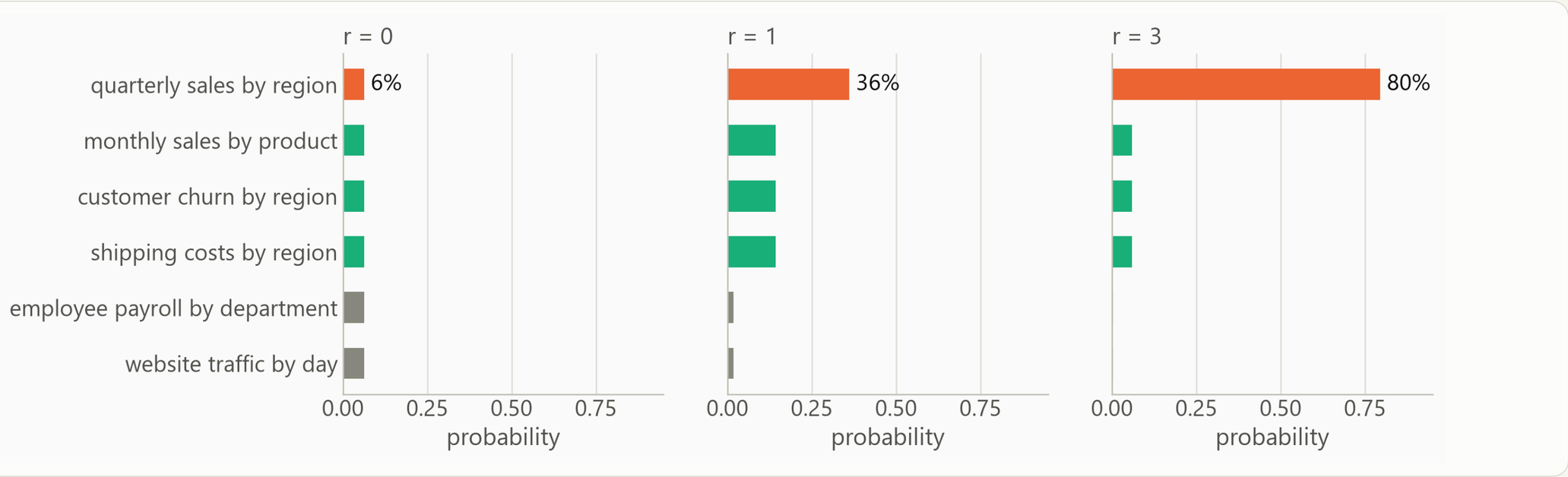
Real chips, no suppression: 5.7σ, 6.8σ and 6.6σ. Plain sampling never passed 2σ

Chart: exact curves, click at step 6. Separation: click at step 3, τ = 3, mean of five snapshots, plain sampling against steer + local.

The other slot: a QRSM as the oracle

Grover has two slots: what to search, and what counts as a match. A **reader QRSM** fills the second. It reads a request token by token, turning the phase of each entry that matches.

"how much did we sell in each area"
→ `[sales, region]`



19 of 20 vague requests found the right table; the one miss was the classical tokeniser

No speedup the oracle is compiled from the catalogue: 32,800 gates against a 1,024-step scan

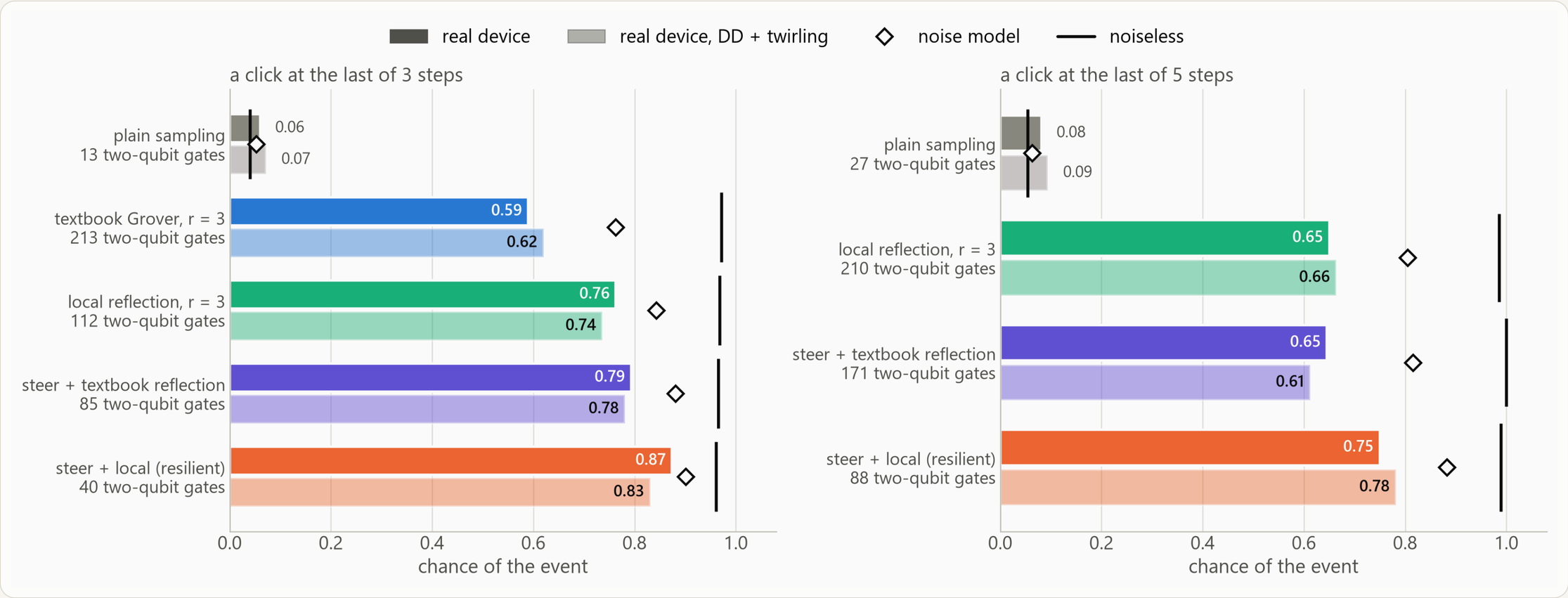
Demo: real circuits, about a minute, no account needed

```
$ python demo/demo.py
A click at step 6: 1,000 shots each
plain sampling                    7.8% usable   distance 0.077
steer + 1 Grover iteration      100.0% usable   distance 0.010
steer + local reflection         96.8% usable   distance 0.054
What follows a click? P(click 1, 2, 3 steps later)
no defect    0.016  0.065  0.096   exact 0.017  0.058  0.101
one defect   0.011  0.027  0.045   exact 0.010  0.030  0.047
“how much did we sell in each area”
quarterly_sales_by_region   81.4%   every entry started at 6.2%
```

Every number comes from a Qiskit circuit run on Aer. `--noisy FakeFez` runs it on an IBM device’s noise model.

The repository also has an interactive lab: watch 48 runs change as you switch builds, and try your own search requests.

On a real IBM chip, fewer gates won



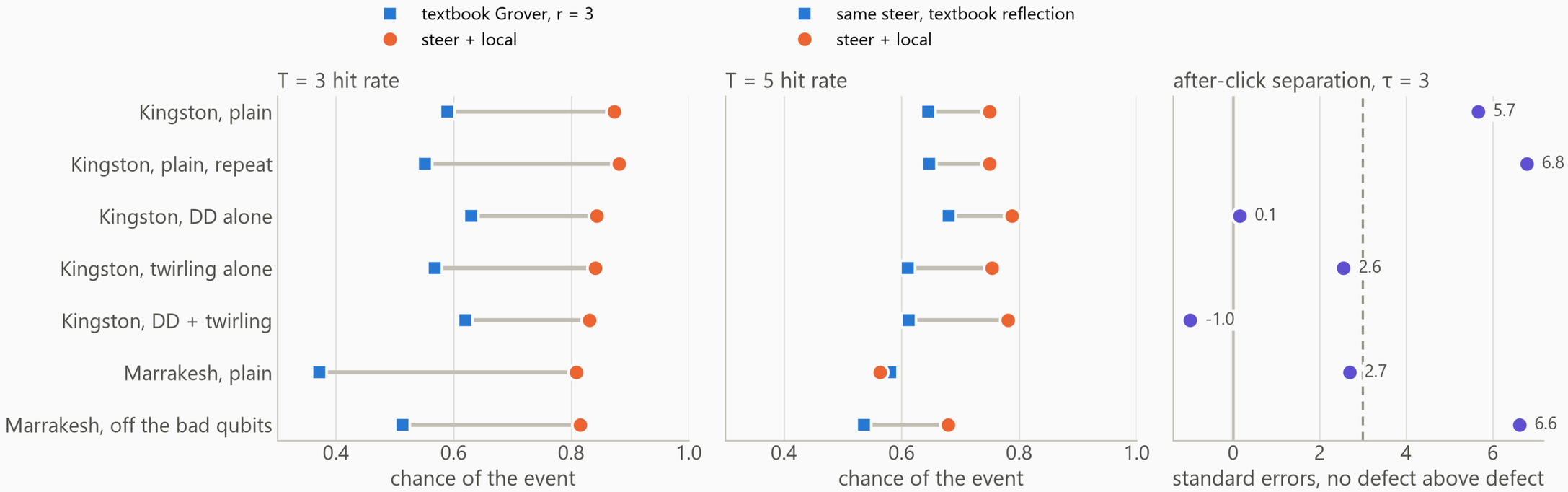
0.87 vs 0.59
the event, on 40 gates against 213

0.06 vs 0.26
distribution error: better answers

20 of 20
comparisons won across five jobs

ibm_kingston, 5 Nov 2026, 4,096 shots, readout-corrected. Jobs dbmfqlt8ft8s738uplmg and dbmfqme74qds73ds7ieg.

What only the real chips could teach us



Depth costs more than modelled

Fidelity fell about **3× faster per gate** than in the device's own noise model: 0.45% against 0.15%.

Suppression isn't free

DD helped textbook a little, and **erased the fingerprint**. We predicted twirling would, and we were wrong.

Calibration can hide a bad patch

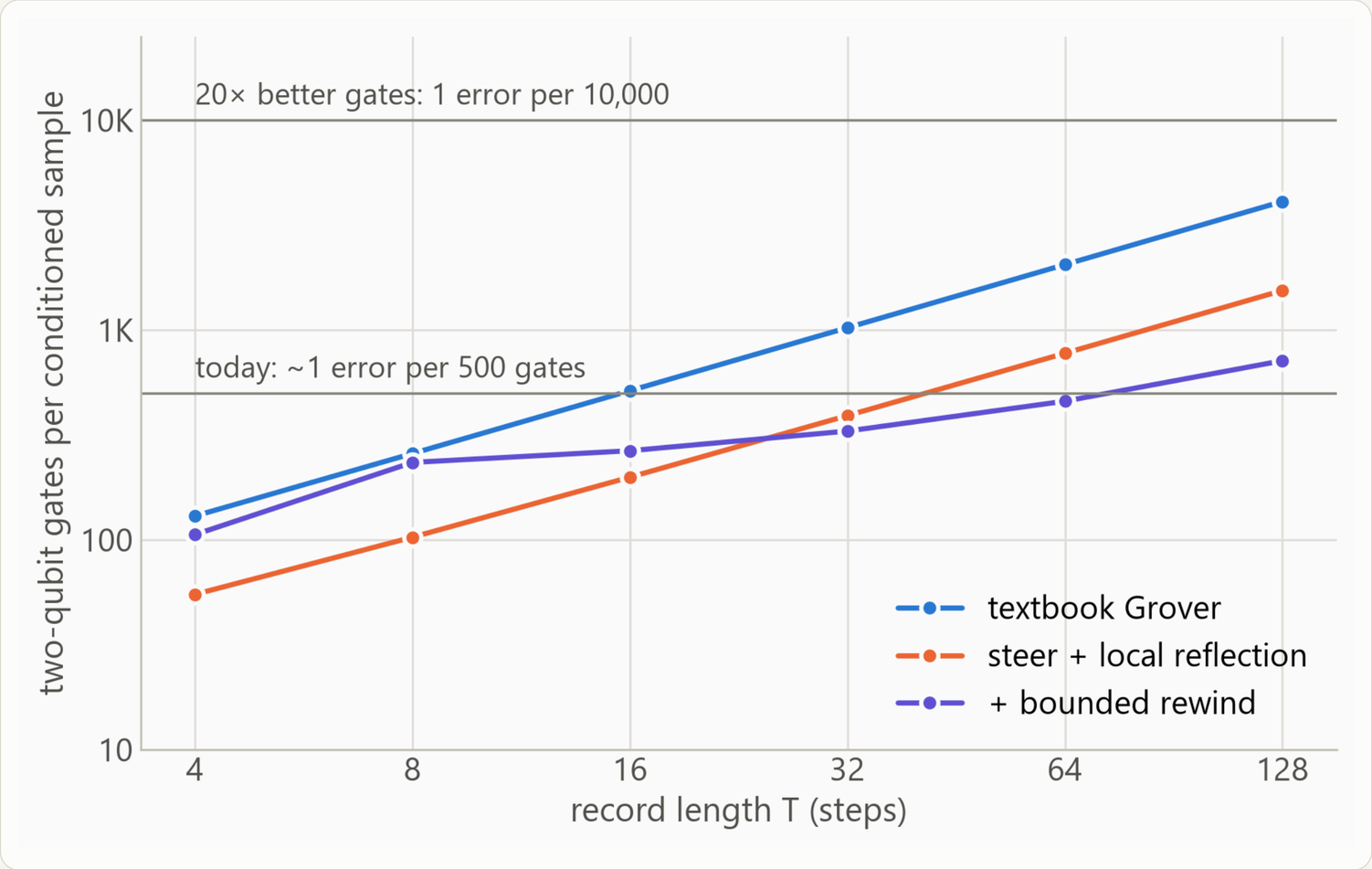
Marrakesh's best-looking qubits were the **worst we used**. Our echoes caught it, and moving off them recovered the result.

Seven jobs, 337 s of device time. Avoiding those qubits was a post-hoc choice, so that job isn't a clean replication.

What we don’t claim

Claim	The honest position
A speedup	None today. An exact classical sampler gives a sample in about a millisecond here. Its cost grows about 6× per hidden qubit, so a crossover is plausible, but not shown.
Rewinding a real chip	We rewind a model of the chip. A physical defect can’t be run backwards, so this queries what was learned from the clicks.
Steering is quantum	It is classical importance sampling. Exact only for clicks at known steps, and it needs a rough value of p.
Local reflection is exact	It is an approximation. Steered, its hit rate falls to 0.73 at T = 14, where the exact build stays at 1.0.
A new algorithm	Amplified rejection sampling is known (Low, Yoder & Chuang 2014). Ours is the mapping onto hidden quantum models, and the resilience work.
Hardware at scale	Seven jobs on two chips in one evening: at most 11 qubits and 5 steps. At 5 steps the distribution error is still 0.14 to 0.21.

Why it has potential: where the classical baseline runs out



A 66× smaller job

Conditioning on a 64-step record: 2.7 million gates as textbook, 40,000 once steered.

Where classical runs out

IBM’s 2029 target, 200 logical qubits and 100 million gates, would fit 50 hidden qubits and 150 steps. That memory alone is about 18 PB classically.

Next

Steering for learned models, error detection, echo-checking layouts first, and rewinding only the memory’s horizon: 8 steps held 0.80 to 0.87 out to T = 18.

Scale estimates, not compiled circuits. Chart: logical gates at p = 0.09. Roadmap: IBM’s published Starling target.

What we learned, and who did what

What we learned

Reversibility is a design choice. One reset decides whether Grover is possible at all.

Exact is not always best. On noisy hardware a cheaper approximate circuit beat the exact one.

Test what you can’t see. The hit rate looked perfect while the hidden state was scrambled, and the calibration favoured a chip’s bad qubits.

The team

Nicholas Taylor: QRSM research, the Grover fusion, experiments

Tim Yudintsev: research, and the keyword search idea

Dan Izquierdo: experimentation

Dillion: formatting and revising

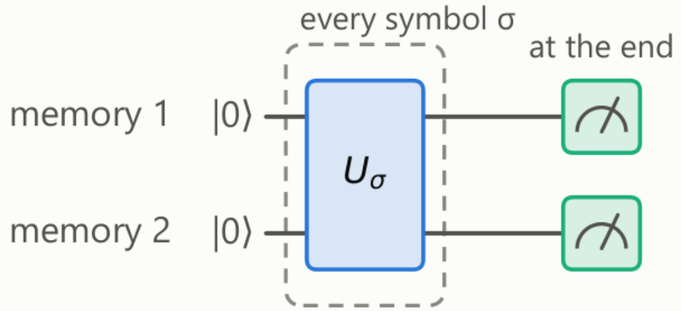
Disclosed

The QRSM model predates the event. The Grover fusion is new this week. Built with AI assistance from Claude.

`github.com/Nicholas55555/rewindandamplify`

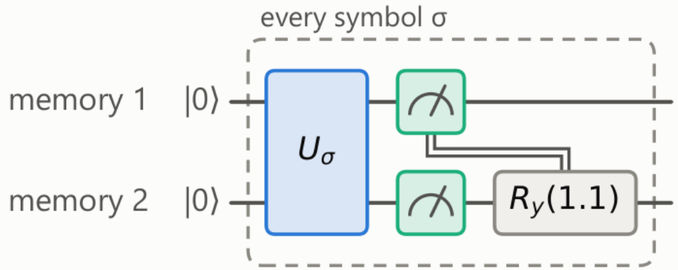
Backup: the QRSM family, and where ours sits

Coherent



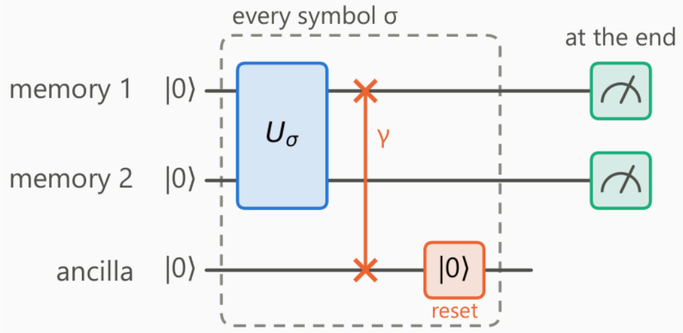
Nothing is reset, so it can’t forget.

Measured



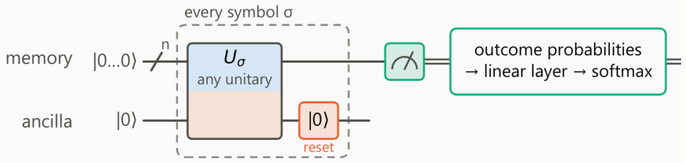
Read every step: a classical automaton.

Hybrid



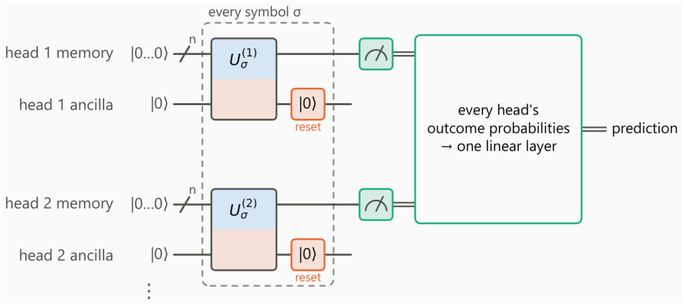
An ancilla carries information away.

Free-unitary hybrid



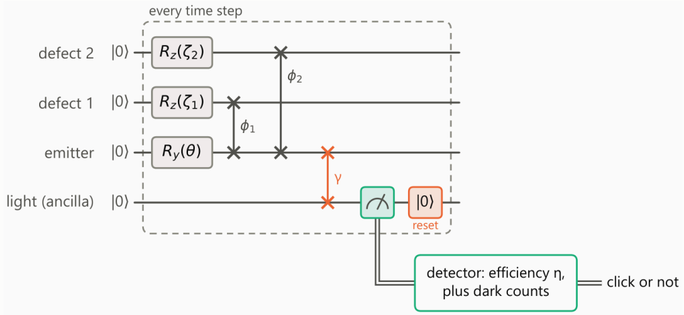
Any unitary, learned for each symbol.

Parallel heads



Small hybrids side by side, never entangled.

Ours: hidden-defect emitter



A physics-shaped hybrid. We run one defect.

From the team’s QRSM circuit diagrams. Grover fit, from our memory-access note: constrained hybrid limited, free-unitary hybrid good for small tests, fully coherent the strongest fit, parallel heads per head only. Purifying makes ours fully coherent.